

APPLYING SUPERVISED LEARNING TO PREDICT STUDENT DROPOUT

[Company name withheld under NDA]

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PROBLEM STATEMENT

Company requested analysis of student's journey data to determine risk factors in student drop out. As well as depriving students of the opportunity to obtain qualifications, students drop out affects profitability. I was tasked with performing in-depth data analysis to uncover risk factors driving student drop-out and to facilitate the design of timely intervention strategies.

DATA STRUCTURE

Student's journey is split into 3 stages: Stage 1/ Applicant and course information Stage 2/ Student engagement data Stage 3/ Academic performance data.

The data also includes categories which would only be available after completion of the study programme (e.g. Destination University), hence the data was used selectively to address the business problem and to prevent data leakage by developing predictions based on information, which would not be available at the time of the prediction being made.

The data has numerical and categorical variables.

dex	Variable Type	Variable Name		data modifications
18	Nominal Categorical	CompletedCourse (Target)	TARGET	
2	Numerical	AssessedModules	STAGE 3 specific	Imputed median values based on x_training data set
3	Numerical	PassedModules	STAGE 3 specific	Imputed median values based on x_training data set
4	Numerical	FailedModules	STAGE 3 specific	Imputed median values based on x_training data set
5	Numerical	AuthorisedAbsenceCount	STAGE 2 specific	Drop missing rows<2%
6	Numerical	UnauthorisedAbsenceCount	STAGE 2 specific	Drop missing rows<2%
0	Numerical	LearnerCode	STAGE 1	
1	Numerical	DateofBirth (converted to Age)	STAGE 1	
7	Ranked Categorical	CourseLevel (Ordinal)	STAGE 1	
8	Nominal Categorical	CentreName	STAGE 1	
9	Nominal Categorical	BookingType	STAGE 1	
10	Nominal Categorical	LeadSource	STAGE 1	
11	Nominal Categorical	DiscountType	STAGE 1	
12	Nominal Categorical	Gender	STAGE 1	
13	Nominal Categorical	Nationality	STAGE 1	
14	Nominal Categorical	HomeState	STAGE 1	>50% missing values removed from the data set
15	Nominal Categorical	HomeCity	STAGE 1	high cardinality; removed together with HomeState
16	Nominal Categorical	CourseName	STAGE 1	
17	Nominal Categorical	IsFirstIntake	STAGE 1	
19	Nominal Categorical	ProgressionDegree		discarded to avoid data leakage
20	Nominal Categorical	ProgressionUniversity		discarded to avoid data leakage

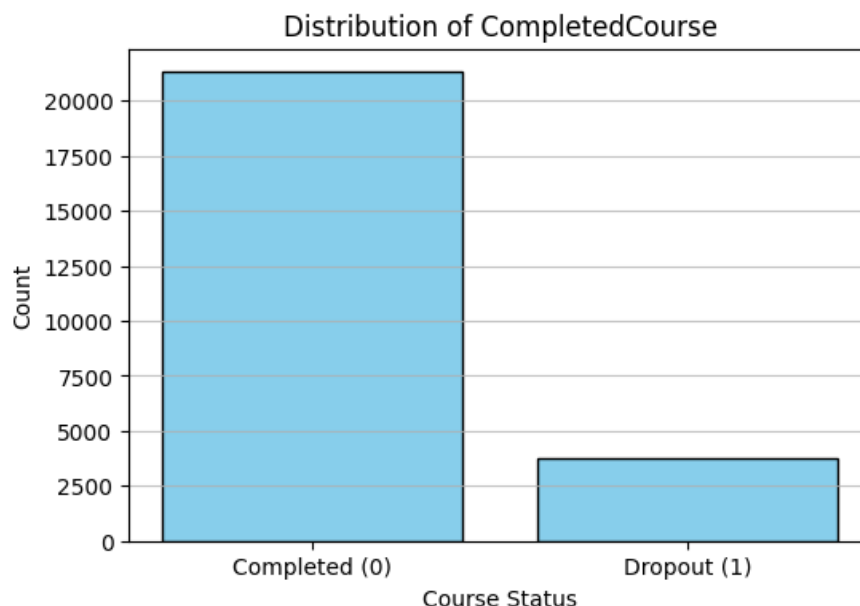
Stage 2 data includes all features used in Stage 1 plus Stage 2 specific features. Stage 3 includes all Stage 2 features plus specific Stage 3 features: *AssessedModules*, *PassedModules*, *FailedModules*.

The features were One Hot Encoded (OHE) with exception of *IsFirstIntake*, *CompletedCourse*, *AssessedModules*, *PassedModules*, *FailedModules*, *AuthorisedAbsenceCount*, *UnauthorisedAbsenceCount*, *Age*.

All Features in the data set are standardised with the exception of the target variable *CompletedCourse*.

DATA IMBALANCE

As it is not unlikely to be observed in the natural phenomena, the drop-out data is highly imbalanced, with 85% students completing the course and 15% dropping out.



This necessitates the use of class weights, thus incentivising the model to focus on the minority class to optimise (drop-outs) predictions.

METHODS

The following methods are applied for customer analysis and customer segmentation.

Method	Key Model Features	Method Outline
Neural Network		
BASE MODEL Different mechanisms of optimisers can be used to define how the Neural Network Learns (ADAM and RMSProp)	Baseline model with standard number of neurons and layers (input, hidden and output layers). Assumptions are made w.r.t number of batches (size of data subsets used in modelling) and epochs (iterations run).	Neural networks are computer systems inspired by human intelligence looking to recognise and model patterns in the data based through deep learning to facilitate decision making. The model's goal is to minimise Loss Function (eg Binary Cross Entropy) through forward

		propagation and back propagation. The embedded optimisers allow to minimise the error by fine-tuning weights and bias within the mathematical functions used to model the data patterns. This is an iterative process performed until “the-best-fit” is found by the model e.g. a global minimum.
• ADAM OPTIMISER	Use of first and second moments: Momentum and Adaptive Learning Rate. Learning Rate is adjusted dynamically for each parameter relative to its weight;	Advantage in handling local saddle points, can change direction of learning and respond to frequent changes in the gradient’s direction. Tracks both first and second moments during optimisation.
• RMSPROP OPTIMISER (Root Mean Square Propagation)	Calculates gradient and exponentially weighted average second moment; tracks second moment only during optimisation.	Component of ADAM Optimiser
XGBoost Model	*Second Order Taylor expansion and Hessian *Lasso 1 and Ridge 1, Learning Rate (eta) regularisation techniques *cross-validation to find optimal n_estimators	Advantageous performance allowing parallel computing High accuracy and robustness Suitable for Gradient Boosting frameworks

EXPERIMENTS AND RESULTS

The analysis was divided into 3 stages to account for data sets availability at different stages of the learning journey.

Following data preprocessing at each respective stage, baseline Neural Network and XGBoost models were built at Stage 1. It was quickly apparent that the models were affected by imbalanced data, which was corrected with class weights for all models. Neural Network models were optimised for a sample number of neurons and layers combinations, with use of two optimiser ADAM and RMSProp. Additionally, the activations functions were set to: RELU for input and hidden layers and Binary Cross Entropy for the Output layers, consistent with the theoretical framework for neural network predictions with binary outcomes (0,1). Additionally, regularisation was introduced to the model to prevent overfitting with use of L2 regularisation and DropOut.

Following the optimisation, a shallow neural network with one hidden layer and 32 neurons was found to perform best for both RMSProp and Adam.

Whilst we can assess the accuracy of the prediction achieved with Neural Networks, we cannot determine the features driving DropOuts. This can be established with XGBoost model.

XGBoost model was optimised for the following parameters grid:

learning_rate: [0.05, 0.15], *max_depth*: [3, 5], *n_estimators*: [100, 200], *subsample*': [0.8, 1.0], *colsample_bytree*': [0.8, 1.0].

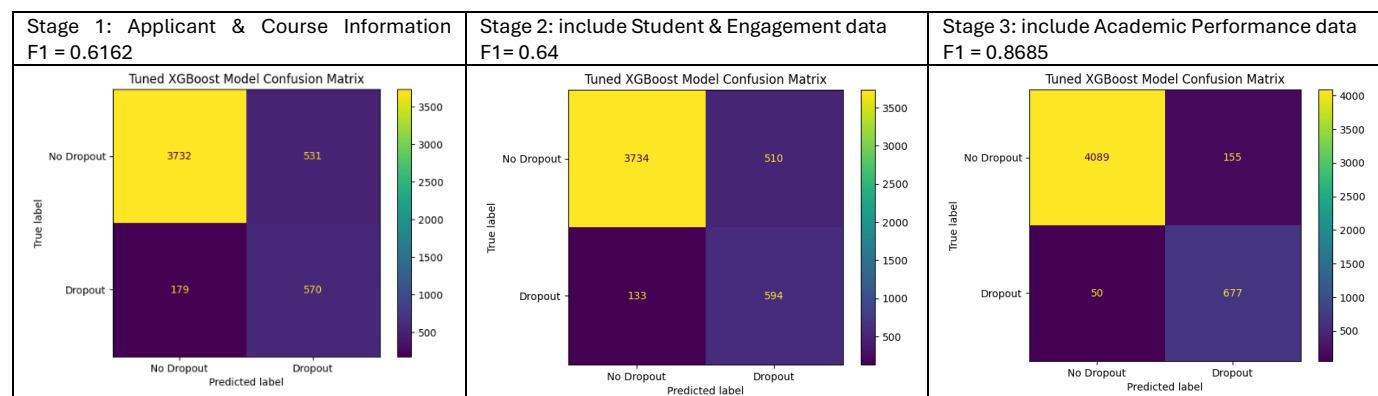
The best performance of the optimised XGBoost model was achieved with parameters

'*colsample_bytree*': 0.8, '*learning_rate*': 0.05, '*max_depth*': 5, '*n_estimators*': 200, '*subsample*': 0.8}.

The best fit of parameters of XGBoost model predicted 531 false positives and 179 false negatives, with F1 = 0.6162.

$F1 = 2 \times (\text{Precision} \times \text{Recall} / \text{Precision} + \text{Recall})$

Overall, XGBoost Tune Model had the best performance across the stages of analysis, decreasing false positives from 531 to 151 and false negatives from 179 to 50 as the data of students & engagement and academic performance was used in model training and validation. As result of tuning, the model performance, as measured by F1, which is the most suitable measure for binary imbalanced data sets, improved from 0.6162 to 0.8695, which is indicative of an excellent model performance.



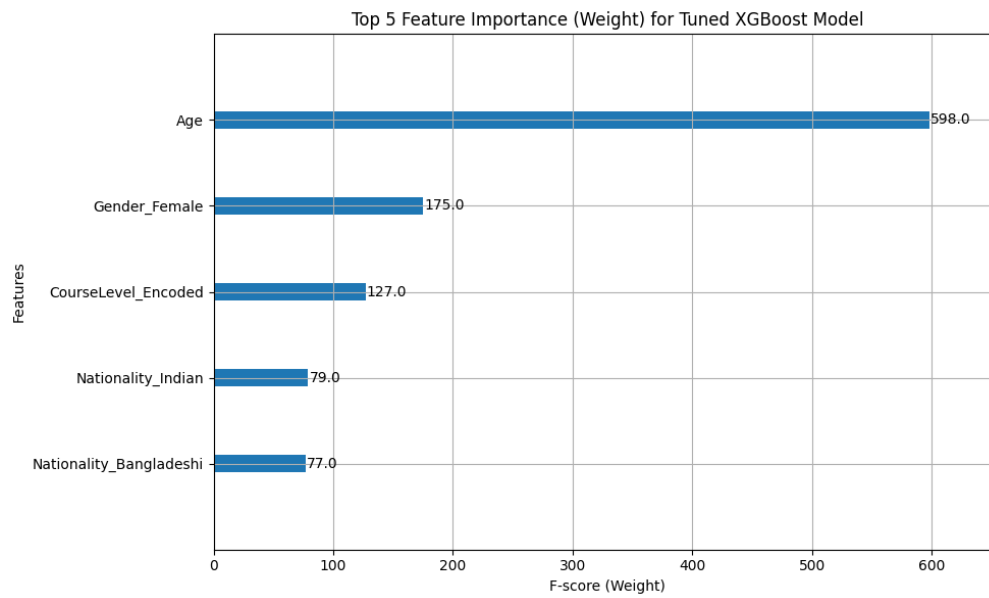
RECOMMENDATIONS

In terms of uncovering management insights, the following factors were most predictive of drop-out at different stages of the academic journey, based on XGBoost Tune model analysis (cf graphs below). At the early stage, the strongest predictor of dropout was **age**, followed by **gender (female)**

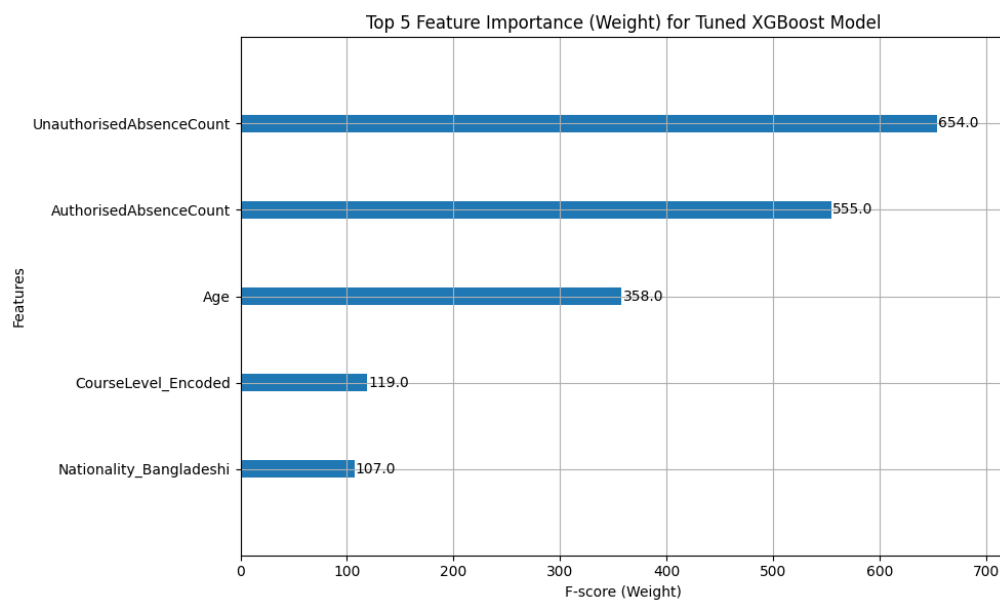
and course level. At Stage 2, the strongest indicators were **unauthorised and authorised absences reflective of student engagement**, and at Stage 3 the factors included: **unauthorised absences, assessed modules and passed modules, authorised absence and age.**

In order to develop a responsive student support system and to maximise the profitability for the Company, it is recommended to develop an early warning system around these specific indicators at each stage of student learning journey to help identify students at risk of drop-out.

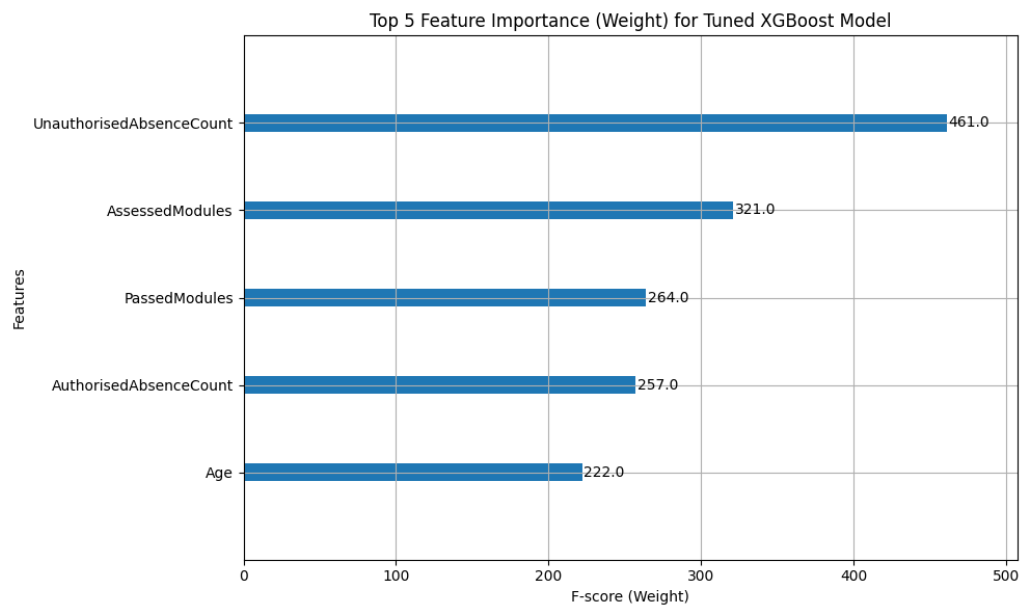
Stage 1: Applicant & Course Information



Stage 2: include Student & Engagement data



Stage 3: include Academic Performance data



APPENDIX

STAGE 1: XGBoost model was also fitted and optimised for a grid of parameters:

learning_rate: [0.05, 0.15], max_depth: [3, 5], n_estimators: [100, 200], subsample: [0.8, 1.0],
colsample_bytree: [0.8, 1.0].

Subsequently, a grid search was performed to establish the best performing parameters combination.

--- Final Model Comparison Summary ---								
	Model Type	Parameters	Train Loss (Logloss)	Validation Loss (Logloss)	Test Accuracy	Test Precision	Test Recall	Test AUC
0	XGBoost (Initial Baseline)	default (no scale_pos_weight)	0.2345	N/A	0.8516	0.7273	0.0107	0.8266
1	XGBoost (Native API Baseline)	eta=0.1, depth=3, scale_pos_weight=5.67	0.3889	0.4089	0.8456	0.4893	0.765	0.8754
2	XGBoost (Tuned)	{'colsample_bytree': 0.8, 'learning_rate': 0.0...	0.3816	0.4025	0.8583	0.5177	0.761	0.8766

XGBoost Tuned – performance indicators

STAGE 1

--- Tuned XGBoost Model Evaluation on Test Set ---

Accuracy: 0.8583

Confusion Matrix:

[[3732 531]

[179 570]]

Precision: 0.5177

Recall: 0.7610

AUC: 0.8766

STAGE 2

--- Tuned XGBoost Model Evaluation on Test Set ---

Accuracy: 0.8706

Confusion Matrix:

[[3734 510]

[133 594]]

Precision: 0.5380

Recall: 0.8171

AUC: 0.9121

STAGE 3

Accuracy: 0.9588

Confusion Matrix:

[[4089 155]

[50 677]]

Precision: 0.8137

Recall: 0.9312

AUC: 0.9901